

Syllabus

Course Code: CS2000		
Name of the Programme: B.Tech(H)		
Name of the Course: Design and Analysis of Algorithms		
Semester: III		
Course credit	No. of hours per week	Total no. of Teaching hours
04	2+1+2	75
Course Objectives:		
Develop proficiency in designing efficient algorithms to solve complex computational problems across diverse domains using techniques like divide-and-conquer, greedy methods, and backtracking.		
Gain the ability to evaluate and compare algorithm performance through rigorous analysis, including time and space complexity, to optimize solutions for practical applications.		
Master the implementation of fundamental algorithms, such as sorting, searching, and graph traversal, in a programming environment to build robust software solutions.		
Cultivate problem-solving skills by applying algorithmic strategies and data structures to real-world challenges, fostering innovation in areas like optimization, networking, and data processing.		

Module - 1 Introduction to Algorithms, Basics of Searching and Sorting	No. of hours
	12
Introduction to Algorithms, Asymptotic notation and growth of functions, Analyzing complexity of algorithms, Recurrence relation, Substitution method, Recursion Tree Method, Master's Theorem, Brute force Design Technique : Linear Search, Naïve string-matching algorithm. Sorting methods: Bubble sort, Insertion Sort, and Selection Sort.	

Module - 2 Divide and Conquer & Decrease and Conquer	No. of hours
	8
Divide & Conquer: Merge Sort, Quick-Sort, Strassen's Matrix Multiplication Decrease & Conquer: Binary Search, Heap sort, Basics of graphs : Graph representation, DFS and BFS, Counting sort.	

Module - 3 Greedy Design Technique	No. of hours
	10
Elements of greedy strategy, Knapsack problem, Job sequencing with deadlines, Huffman encoding. Spanning trees, Minimum cost spanning trees: Kruskal's and Prim's algorithms; Single source shortest path algorithms, Dijkstra's Algorithms.	

Module - 4 Dynamic Programming	No. of hours
	8
Elements of dynamic programming, illustration of use of dynamic programming with specific examples such as 0/1 Knapsack problem, Matrix chain multiplication, finding longest common subsequence, Bellman-Ford Algorithm, Floyd Warshal's algorithm.	

Module - 5 String Matching and NP Completeness	No. of hours
	7
Backtracking: N-Queens problem, Hamiltonian Cycle. Branch & Bound: Knapsack Problem String matching: Rabin-Karp algorithm, Knuth-Morris-Pratt algorithm. Introduction of NP-completeness: P, NP, NPC, NPH problems.	

Lab Programs		[30 Hours]
1	Program to implement Binary Search using Divide and Conquer	
2	Program to implement minimum and maximum element in given array using Divide and Conquer	
3	Program to implement Merge sort using Divide and Conquer	
5	Program to implement Quick sort	
6	Program to implement Heap sort	
7	Program to implement Graph Traversal: Breadth First Traversal	
8	Program to implement Graph Traversal: Depth First Traversal	
9	Program to implement Knapsack problem using Greedy method	
10	Program to implement Prim's algorithm using Greedy method	
11	Program to implement Kruskal's algorithm using Greedy method	
12	Program to implement All Pairs Shortest Path Using Dynamic Programming	

Course Outcomes: After completing the course, the students will be able to:	
CO1	Understand and analyze algorithms
CO2	Design efficient algorithms using Divide & Conquer and Decrease & Conquer techniques
CO3	Apply greedy & dynamic programming strategies to optimization problems
CO4	Explore advanced algorithm design techniques

Text books	
1.	Cormen H. T., Leiserson E. C., Rivest L. R., and Stein C., Introduction to Algorithms, MIT Press (2009) 3rd ed. ISBN 978-0-262-53305-8
2.	Anany Leviton, "Introduction to the Design and Analysis of Algorithms", 3rd Edition, ISBN 13: 978-0-13-231681-1.

Reference books

1.	Aho, Hopcraft, Ullman, “The Design and Analysis of Computer Algorithms”, Pearson [ISBN-10 : 0201000296, ISBN-13 : 978-0201000290]
2.	Dasgupta, Papadimitriou and Vazirani, “Algorithms”, TMH [ISBN-10 : 9780073523408, ISBN-13 : 978-0073523408]
3.	Complete YouTube Playlist (Hindi) by Shoeb Ahmad (Subject Teacher) Design and Analysis of Algorithms: https://shorturl.at/Ryz2L